

CHAPTER 5

THE VAPNIK-CHERVONENKIS PROPERTY IN O-MINIMAL STRUCTURES

Introduction

The main result in this chapter is as follows, see (3.15) below.

THEOREM. *Let $(R, <, \mathcal{S})$ be an o-minimal structure and $S \subseteq R^{p+q}$ a definable set. Then there is a positive integer $d = d(S)$ such that for all sufficiently large $n \in \mathbb{N}$ each n -element set $F \subseteq R^q$ has at most n^d subsets of the form $S_x \cap F$ with $x \in R^p$.*

This exhibits in a purely combinatorial way that the variation among the sets S_x is highly restricted as x ranges over the parameter space R^p : the total number of subsets of an n -element set is 2^n , and this grows much faster with n than the polynomial function n^d . The key step (taken in Section 3) is to show that a model-theoretic structure $\mathcal{R} = (R, \dots)$ has this combinatorial property, if it has the property for $q = 1$ (Shelah). The case $q = 1$ is easily checked for o-minimal structures, see (2.7) and (2.11). We also derive some other results of a rather general nature.

The combinatorial fact expressed by our theorem has an interpretation in terms of probability; see the notes at the end of the chapter. The material in this chapter will not be used later on, but it would be a pity to omit it. Moreover, I believe there is ample room for further developments along the lines of this chapter.

§1. A combinatorial dichotomy

(1.1) Let $\mathcal{C}$ be a collection of subsets of an infinite set X . Given $F \subseteq X$ we put

$$C \cap F := \{C \cap F : C \in \mathcal{C}\},$$

the set of intersections of sets in $\mathcal{C}$ with F . If $A \subseteq F$ is of the form $A = C \cap F$ for some $C \in \mathcal{C}$ we also say that **A is cut out from F by a set in $\mathcal{C}$** . So $|C \cap F| \leq 2^{|F|}$, where $|S|$ denotes the cardinality of a set S . Now we define a function $f_{\mathcal{C}} : \mathbb{N} \rightarrow \mathbb{N}$ that in some sense measures the complexity of the collection $\mathcal{C}$:

$$f_{\mathcal{C}}(n) := \max\{|C \cap F| : F \text{ is an } n\text{-element subset of } X\}.$$

So $0 \leq f_{\mathcal{C}}(n) \leq 2^n$ for all n . We have the following, perhaps surprising, dichotomy.

(1.2) THEOREM. *Either $f_{\mathcal{C}}(n) = 2^n$ for all n , or else there is $d \in \mathbb{N}$ such that $f_{\mathcal{C}}(n) \leq n^d$ for all sufficiently large n .*

The first possibility means that for each n there is an n -element set $F \subseteq X$ all of whose subsets are cut out from F by sets in $\mathcal{C}$, while in the second case for all sufficiently large finite sets $F \subseteq X$ relatively few subsets of F are cut out from F by sets in $\mathcal{C}$. This theorem will be derived from the following combinatorial lemma:

(1.3) LEMMA. *Let F be a finite set of size n and $\mathcal{D} \subseteq \mathcal{P}(F)$, a collection of subsets of F , and suppose that*

$$|\mathcal{D}| > \sum_{i \leq d} \binom{n}{i},$$

where $0 \leq d \leq n$. Then F has a subset E such that $|E| = d$ and $\mathcal{D} \cap E = \mathcal{P}(E)$.

(1.4) REMARKS.

1. The hypothesis is sharp since the collection of subsets of F of size $< d$ has cardinality equal to the indicated sum of binomial coefficients, and this particular collection violates the conclusion of the lemma.

2. Let us write the indicated sum of binomial coefficients as $p_d(n)$. Clearly there is a unique polynomial $p_d(X) \in \mathbb{Q}[X]$ (of degree $d-1$ if $d \geq 1$, $p_0(X) = 0$), whose value at n is $p_d(n)$ for $n \geq d$. Note that then $p_d(x)$ is defined for arbitrary real x , and the ‘‘Pascal triangle equality’’ gives

$$p_{d-1}(x-1) + p_d(x-1) = p_d(x) \quad (d \geq 1).$$

(1.5) PROOF OF THE COMBINATORIAL LEMMA. By induction on n . The desired result trivially holds for $d = 0$ and $d = n$, so let $0 < d < n$. Pick a point $x \in F$ and let $F' := F - \{x\}$. Also put $D' := D - \{x\}$ for $D \in \mathcal{D}$, and let $\mathcal{D}' := \{D' : D \in \mathcal{D}\}$. Note that under the map $D \mapsto D' : \mathcal{D} \rightarrow \mathcal{D}'$ a set $S \in \mathcal{D}'$ has either exactly one preimage or exactly two preimages; in the latter case these two preimages are S and $S \cup \{x\}$. So $\mathcal{D}' = \mathcal{D}_1 \cup \mathcal{D}_2$ (a disjoint union) where $\mathcal{D}_1$ contains those $S \in \mathcal{D}'$ having one preimage in $\mathcal{D}$ and $\mathcal{D}_2$ those with two preimages. If $|\mathcal{D}'| > p_d(n-1)$ then by the inductive assumption applied to F' and $\mathcal{D}'$ there is a subset E of F' of size d with $\mathcal{D}' \cap E = \mathcal{P}(E)$, hence $\mathcal{D} \cap E = \mathcal{P}(E)$ and we are done.

So assume $|\mathcal{D}'| \leq p_d(n-1)$. But

$$\begin{aligned} |\mathcal{D}| &= |\mathcal{D}_1| + 2|\mathcal{D}_2| = (|\mathcal{D}_1| + |\mathcal{D}_2|) + |\mathcal{D}_2| \\ &= |\mathcal{D}'| + |\mathcal{D}_2| > p_d(n) = p_d(n-1) + p_{d-1}(n-1), \end{aligned}$$

hence $|\mathcal{D}_2| > p_{d-1}(n-1)$, so again by the inductive assumption applied to F' and $\mathcal{D}_2$ there is a set $E \subseteq F'$ of size $d-1$ such that $\mathcal{D}_2 \cap E = \mathcal{P}(E)$. Since for each set $S \in \mathcal{D}_2$ we have $S \in \mathcal{D}$ and $S \cup \{x\} \in \mathcal{D}$ this gives $\mathcal{D} \cap (E \cup \{x\}) = \mathcal{P}(E \cup \{x\})$. $\square$

We now prove theorem (1.2) in the following more precise form.

(1.6) THEOREM. *Suppose d is a nonnegative integer with $f_C(d) < 2^d$. Then $f_C(n) \leq p_d(n)$ for all n .*

PROOF. If $n < d$ then $p_d(n) = 2^n$ and the desired inequality holds trivially. So let $n \geq d$, and let F be an n -element subset of X . If $|\mathcal{C} \cap F| > p_d(n)$, then the lemma would give us a set $E \subseteq F$ of size d such that $|\mathcal{C} \cap E| = 2^d$, contradicting the assumption $f_C(d) < 2^d$. Hence $|\mathcal{C} \cap F| \leq p_d(n)$. Since F was arbitrary, the desired result follows. $\square$

REMARK. The inequalities in the theorem are sharp, since the collection $\mathcal{C}$ of subsets of X of size $< d$ satisfies $f_C(d) < 2^d$, $f_C(e) = 2^e$ for $e < d$, and $f_C(n) = p_d(n)$ for all n .

§2. Vapnik-Chervonenkis classes and dependence

(2.1) Given an infinite set X , a collection $\mathcal{C} \subseteq \mathcal{P}(X)$ is said to be a **Vapnik-Chervonenkis class** (or VC-class) if $f_C(d) < 2^d$ for some $d \in \mathbb{N}$, and then we define its **Vapnik-Chervonenkis index** $V(\mathcal{C})$ to be the least such d ; if $\mathcal{C}$ is not a VC-class we put $V(\mathcal{C}) = \infty$. By theorem (1.6), $V(\mathcal{C}) < \infty$ means that f_C is of polynomial growth, while $V(\mathcal{C}) = \infty$ means that $f_C(n) = 2^n$ for all n .

(2.2) Usually a collection $\mathcal{C} \subseteq \mathcal{P}(X)$ will be indexed by elements of a parameter space or index set, and in fact the set X and this index set play dual roles. To bring out this duality, let us assume that infinite sets X and Y are given, and a binary relation $\Phi \subseteq X \times Y$. For $x \in X$ and $y \in Y$ we put

$$\Phi_x := \{y \in Y : (x, y) \in \Phi\}, \quad \Phi^y := \{x \in X : (x, y) \in \Phi\}.$$

Let $\Phi^Y := \{\Phi^y : y \in Y\} \subseteq \mathcal{P}(X)$ and $\Phi_X := \{\Phi_x : x \in X\} \subseteq \mathcal{P}(Y)$. (So Y is the parameter space for Φ^Y and X is the parameter space for Φ_X .)

Let $E \subseteq F \subseteq X$. Then we have for $y \in Y$ the obvious equivalence

$$E = \Phi^y \cap F \Leftrightarrow y \in \Phi_x \text{ for all } x \in E \text{ and } y \notin \Phi_x \text{ for all } x \in F - E,$$

hence

$$(*) \quad E \in \Phi^Y \cap F \Leftrightarrow \left(\bigcap_{x \in E} \Phi_x \right) \cap \left(\bigcap_{x \in F-E} (Y - \Phi_x) \right) \neq \emptyset.$$

As E ranges over the subsets of a finite set $F \subseteq X$, the nonempty intersections on the right in $(*)$ are exactly the atoms of the boolean algebra of subsets of Y generated by the Φ_x with $x \in F$. So we conclude (for finite $F \subseteq X$):

$$(**) \quad \begin{cases} |\Phi^Y \cap F| = \text{number of atoms of the boolean algebra of} \\ \text{subsets of } Y \text{ generated by the sets } \Phi_x \text{ with } x \in F. \end{cases}$$

(2.3) This leads us to a situation dual to the earlier set-up: Fix a set Y . For $S \subseteq Y$ we write $S^1 := S$, and $S^{-1} := Y - S$ (the complement of S in Y). Given sets $S_1, \dots, S_n \subseteq Y$ we note that the 2^n sets (many of which may be empty)

$$S_1^{\epsilon(1)} \cap \dots \cap S_n^{\epsilon(n)}, \quad \epsilon: \{1, \dots, n\} \rightarrow \{-1, 1\},$$

are disjoint and cover Y , so the boolean algebra $B(S_1, \dots, S_n)$ of subsets of Y generated by $S_1, \dots, S_n$ has as its atoms the *nonempty* sets among these intersections. Hence the maximal number of atoms is 2^n and this is achieved precisely when all 2^n intersections indicated are nonempty. In this case we say that the sequence $S_1, \dots, S_n$ is **independent** (in Y), or (abusing language) that $S_1, \dots, S_n$ are independent (in Y). Otherwise we say that the sequence $S_1, \dots, S_n$ is **dependent**. We also say that a collection $\mathcal{G}$ of subsets of Y is **independent** (in Y) if for each $n \in \mathbb{N}$ there is an independent sequence $S_1, \dots, S_n$ in $\mathcal{G}$, and otherwise we call $\mathcal{G}$ **dependent**. The following is now dual to theorem (1.6).

(2.4) PROPOSITION. *Suppose $\mathcal{G} \subseteq \mathcal{P}(Y)$ is dependent, so there is d such that each sequence $S_1, \dots, S_d$ in $\mathcal{G}$ is dependent. Then for such d and all $S_1, \dots, S_n \in \mathcal{G}$, the boolean algebra $B(S_1, \dots, S_n)$ has at most $p_d(n)$ atoms.*

PROOF. Let $n \geq d$ and $S_1, \dots, S_n \in \mathcal{G}$. Let $F := \{1, \dots, n\}$, and let $\mathcal{D} \subseteq \mathcal{P}(F)$ consist of the sets $D \subseteq F$ for which the intersection

$$\left(\bigcap_{i \in D} S_i \right) \cap \left(\bigcap_{i \notin D} S_i^{-1} \right)$$

is nonempty. If $|\mathcal{D}| > p_d(n)$ then by lemma (1.3) there would be a set $E \subseteq \{1, \dots, n\}$ of size d such that $\mathcal{P}(E) = \mathcal{D} \cap E$, hence the boolean algebra generated by the sets S_i with $i \in E$ would have 2^d atoms, contradicting the assumption on d . $\square$

(2.5) The simplest examples of dependent collections are obtained as follows. Let $\mathcal{B}$ be a collection of subsets of Y and d a positive integer such that each nonempty intersection $B_1 \cap \dots \cap B_d$ of d sets from $\mathcal{B}$ equals an intersection $\bigcap_{i \in D} B_i$ for some proper subset D of the index set $\{1, \dots, d\}$. For instance, this is the case for $\mathcal{B} =$ the set of connected subsets of the real line $\mathbb{R}$, with $d = 3$. More generally, given an o-minimal structure $(R, <, \mathcal{S})$, we take $Y = R$ and $\mathcal{B} =$ the collection of definably connected subsets of R ; then $\mathcal{B}$ satisfies the property above with $d = 3$, as is easily checked.

(2.6) LEMMA. *Let $\mathcal{B}$ and d be as in the general statement of (2.5). Let $\mathcal{G} \subseteq \mathcal{P}(Y)$ and suppose $e \in \mathbb{N}$ is such that each set in $\mathcal{G}$ is a boolean combination of at most e sets in $\mathcal{B}$. Then $\mathcal{G}$ is dependent. More precisely, $B(S_1, \dots, S_n)$ has at most $p_d(en)$ atoms, for all $S_1, \dots, S_n \in \mathcal{G}$.*

PROOF. Let each S_i be a boolean combination of sets $B_{i1}, \dots, B_{ie}$ in $\mathcal{B}$, and let $F := \{(i, j) : 1 \leq i \leq n, 1 \leq j \leq e\}$, an index set of en elements. Then the boolean

algebra generated by the S_i 's is contained in the boolean algebra generated by the B_λ 's with $\lambda \in F$, and the latter boolean algebra has as atoms the nonempty sets among the intersections

$$\left(\bigcap_{\lambda \in D} B_\lambda \right) \cap \left(\bigcap \left\{ B_\mu^{-1} : \mu \in F, B_\mu \text{ does not contain } \bigcap_{\lambda \in D} B_\lambda \right\} \right)$$

where D varies over the subsets of F of size $< d$. Hence there are at most $p_d(en)$ atoms in the boolean algebra generated by the B_λ 's with $\lambda \in F$, and hence at most $p_d(en)$ atoms in the boolean algebra generated by $S_1, \dots, S_n$. $\square$

(2.7) **EXAMPLE.** Let $(R, <, \mathcal{S})$ be an o-minimal structure and $\Phi \subseteq R^{m+1}$ a definable set and put $\mathcal{G} := \{\Phi_x : x \in R^m\}$, a collection of subsets of R . By Chapter 3, (3.6) we know that for some fixed $e \in \mathbb{N}$ each set in $\mathcal{G}$ has at most e definably connected components. Hence by taking for $\mathcal{B}$ the collection of definably connected subsets of R the lemma shows that $\mathcal{G}$ is dependent, and that the boolean algebra generated by sets $S_1, \dots, S_n \in \mathcal{G}$ has at most $p_3(en)$ atoms. Later we shall see that for each definable set $\Phi \subseteq R^{m+n}$ the collection $\Phi_X := \{\Phi_x : x \in R^m\}$ of subsets of R^n is dependent.

(2.8) Given an infinite set Y and a collection $\mathcal{G} \subseteq \mathcal{P}(Y)$ we define a "growth function" $f^{\mathcal{G}} : \mathbb{N} \rightarrow \mathbb{N}$ by

$$f^{\mathcal{G}}(n) := \text{maximum over all } S_1, \dots, S_n \in \mathcal{G} \text{ of the} \\ \text{number of atoms of } B(S_1, \dots, S_n).$$

In the case where $\mathcal{G}$ is dependent, we let $D(\mathcal{G})$ be the smallest $d \in \mathbb{N}$ for which $f^{\mathcal{G}}(d) < 2^d$, and we call $D(\mathcal{G})$ the **dependency index** of $\mathcal{G}$. If $\mathcal{G}$ is independent we set $D(\mathcal{G}) = \infty$.

(2.9) Suppose X and Y are infinite sets and $\Phi \subseteq X \times Y$ is a binary relation.

Then we defined in (2.2) above the collections $\mathcal{C} = \Phi^Y \subseteq \mathcal{P}(X)$ and $\mathcal{G} = \Phi_X \subseteq \mathcal{P}(Y)$, and (*) and (**) show that $f_{\mathcal{C}} = f^{\mathcal{G}}$, hence $V(\Phi^Y) = D(\Phi_X)$, and in particular,

$$\Phi^Y \text{ is a VC-class} \Leftrightarrow \Phi_X \text{ is dependent,}$$

and by reversing the roles of $\mathcal{C}$ and $\mathcal{G}$,

$$\Phi^Y \text{ is dependent} \Leftrightarrow \Phi_X \text{ is a VC-class.}$$

Less obvious perhaps is

(2.10) **PROPOSITION.** Φ^Y is independent if and only if Φ_X is independent.

PROOF. By symmetry we need only prove one direction, so assume Φ^Y is independent, and let $n \in \mathbb{N}$. We want to find $x(1), \dots, x(n) \in X$ such that $\Phi_{x(1)}, \dots, \Phi_{x(n)} \subseteq$

Y are independent. We know there are 2^n independent sets $\Phi^{y(\epsilon)}$ ($\epsilon \in \{-1, 1\}^n$) in X , so in particular there is for each $m \in \{1, \dots, n\}$ a point $x(m)$ in X with

$$x(m) \in \left(\bigcap_{\epsilon(m)=1} \Phi^{y(\epsilon)} \right) \cap \left(\bigcap_{\epsilon(m)=-1} (\Phi^{y(\epsilon)})^{-1} \right),$$

that is, $(x(m), y(\epsilon)) \in \Phi \Leftrightarrow \epsilon(m) = 1$, for all $\epsilon \in \{1, -1\}^n$, hence

$$y(\epsilon) \in \Phi_{x(m)} \Leftrightarrow \epsilon(m) = 1,$$

and thus

$$y(\epsilon) \in \bigcap_{1 \leq m \leq n} (\Phi_{x(m)})^{\epsilon(m)}.$$

Therefore $\Phi_{x(1)}, \dots, \Phi_{x(n)}$ are independent. $\square$

(2.11) Because of this proposition and the equivalences in (2.9) the properties “ Φ^Y is a VC-class”, “ Φ^Y is dependent”, “ Φ_X is a VC-class”, “ Φ_X is dependent” are all equivalent. The proof of (2.10) shows moreover

$$D(\Phi^Y) > 2^n \Rightarrow D(\Phi_X) > n,$$

hence

$$D(\Phi_X) \leq n \Rightarrow D(\Phi^Y) \leq 2^n,$$

or in terms of $\mathcal{C} = \Phi^Y$ alone,

$$D(\mathcal{C}) \leq 2^{V(\mathcal{C})}.$$

We call the binary relation Φ **dependent** or **independent**, according to whether Φ_X is dependent or independent, and we set $D(\Phi) := D(\Phi_X) = V(\Phi^Y)$.

A useful source of VC-classes is the following result from Dudley [24].

(2.12) PROPOSITION. *Let $\mathcal{L}$ be an m -dimensional real vector space of real-valued functions on an infinite set X , and for each $f \in \mathcal{L}$, put $\text{pos}(f) := \{x \in X : f(x) > 0\}$. Then $\text{pos}(\mathcal{L}) := \{\text{pos}(f) : f \in \mathcal{L}\}$ is a VC-class of subsets of X with $V(\text{pos}(\mathcal{L})) = m + 1$.*

PROOF. Let $A \subseteq X$ be a set with $|A| = m + 1$. We shall see that then not each subset of A is cut out from A by a set in $\text{pos}(\mathcal{L})$. The restriction map $f \mapsto f|_A : \mathcal{L} \rightarrow \mathbb{R}^A$ cannot be surjective since $\dim(\mathbb{R}^A) = m + 1 > \dim(\mathcal{L})$. Therefore $\mathbb{R}^A$ contains a nonzero vector w that is orthogonal to all $f|_A$ ($f \in \mathcal{L}$) with respect to the standard inner product $(u, v) := \sum_{a \in A} u(a) \cdot v(a)$ on $\mathbb{R}^A$. Replacing w by $-w$ if necessary we may as well assume that $A^+ := \{a \in A : w(a) > 0\}$ is nonempty. If there were $f \in \mathcal{L}$ with $A^+ = A \cap \text{pos}(f)$, we would have $0 = (w, f|_A) = \sum w(a) \cdot f(a) > 0$, a contradiction. Hence A^+ is not cut out from A by any set in $\text{pos}(\mathcal{L})$. This shows

$V(\text{pos}(\mathcal{L})) \leq m + 1$, and we leave the proof that $V(\text{pos}(\mathcal{L}))$ is not less than $m + 1$ as an exercise. $\square$

(2.13) In particular the proposition applies to the vector space of real polynomial functions on $X = \mathbb{R}^N$ of degree $\leq d$, where N is a positive integer. In the exercises below this leads to a quite elementary proof that every semialgebraic binary relation $\Phi \subseteq \mathbb{R}^M \times \mathbb{R}^N$ ($M, N > 0$) is dependent. (This is also a special case of the main theorem of this chapter.)

(2.14) EXERCISES.

1. Finish the proof of (2.12) by showing that $V(\text{pos}(\mathcal{L})) \geq m + 1$.

Let X, Y be infinite sets. Given $\Phi \subseteq X \times Y$, put $f^\Phi := f^\mathcal{G} : \mathbb{N} \rightarrow \mathbb{N}$, where $\mathcal{G} = \Phi_X$, so $f^\Phi(n)$ is the maximum over all $x(1), \dots, x(n)$ in X of the number of atoms of the boolean algebra $B(\Phi_{x(1)}, \dots, \Phi_{x(n)})$.

Let $\Phi, \Psi \subseteq X \times Y$, and define the relations $\neg\Phi, \Phi \vee \Psi, \Phi \& \Psi \subseteq X \times Y$ as follows: $(x, y) \in \neg\Phi$ iff $(x, y) \notin \Phi$, $(x, y) \in \Phi \vee \Psi$ iff $(x, y) \in \Phi$ or $(x, y) \in \Psi$, $(x, y) \in \Phi \& \Psi$ iff $(x, y) \in \Phi$ and $(x, y) \in \Psi$. Note that $f^\Phi = f^{\neg\Phi}$.

2. Show that $f^{\Phi \vee \Psi} \leq f^\Phi \cdot f^\Psi$, and derive that $f^{\Phi \& \Psi} \leq f^\Phi \cdot f^\Psi$.

3. Show that if Φ and Ψ are dependent, then $\Phi \vee \Psi$ and $\Phi \& \Psi$ are dependent.

4. Derive from the previous two exercises and proposition (2.12) that every semi-algebraic relation $\Phi \subseteq \mathbb{R}^M \times \mathbb{R}^N$ ($M, N > 0$) is dependent.

§3. Reduction to the case $q = 1$

In this section we prove the following rather difficult result due to Shelah [53]:

(3.1) THEOREM. *Let $\mathcal{R} = (R, \dots)$ be an infinite model-theoretic structure and suppose all definable relations $\Phi \subseteq R^p \times R$, for all $p > 0$, are dependent. Then all definable relations $\Phi \subseteq R^p \times R^q$, for all $p, q > 0$, are dependent.*

(3.2) Following Laskowski [38] we establish this in a purely combinatorial way, via a somewhat more precise result than (3.1), namely theorem (3.12) below. As an essential tool we first prove a well-known combinatorial fact due to Ramsey.

(3.3) Given a set X and a positive integer r , we put

$$X^{(r)} := \text{the collection of all } r\text{-element subsets of } X.$$

(3.4) RAMSEY'S THEOREM. *Given positive integers M, r and k there is a positive integer $N = N(M, r, k)$ so large that if X is a set with $|X| \geq N$ and $X^{(r)} = P_1 \cup P_2 \cup \dots \cup P_k$, then there is a set $Y \subseteq X$ with $|Y| = M$ such that $Y^{(r)} \subseteq P_j$ for some $j \in \{1, \dots, k\}$.*

PROOF. Clearly it suffices to prove this for $k = 2$, and we do this case by induction on r . If $r = 1$ we can take $N = 2M$. Assume true for a certain r , so we have positive integers $N(M, r, 2)$ with the desired property. Next define positive integers $N(1), \dots, N(2M)$ by descending recursion:

$$N(2M) = 1 \text{ and } N(i) := N(N(i+1), r, 2) + 1 \text{ for } 1 \leq i < 2M.$$

We show that then $N = N(M, r+1, 2) := N(1)$ has the required property. Let $X = \{1, \dots, N\}$, with $N = N(1)$ as above, and let $X^{(r+1)} = P_1 \cup P_2$. We construct a descending sequence $A_1 \supseteq A_2 \supseteq \dots \supseteq A_{2M}$ of subsets of X , with $|A_i| \geq N(i)$, and a sequence $a_1, \dots, a_{2M}$, with $a_i \in A_i$ for $i = 1, \dots, 2M$, and $a_i \notin A_{i+1}$ for $i < 2M$. To start with, put $A_1 = X = \{1, \dots, N\}$, and let a_1 be any element of A_1 . Suppose A_i and $a_i \in A_i$ have been constructed, $|A_i| \geq N(i)$, $1 \leq i < 2M$. Then $(A_i - \{a_i\})^{(r)} = Q_1 \cup Q_2$, with

$$Q_1 := \{\{b_1, \dots, b_r\} \in (A_i - \{a_i\})^{(r)} : \{a_i, b_1, \dots, b_r\} \in P_1\},$$

$$Q_2 := \{\{b_1, \dots, b_r\} \in (A_i - \{a_i\})^{(r)} : \{a_i, b_1, \dots, b_r\} \in P_2\}.$$

Since $|A_i - \{a_i\}| \geq N(N(i+1), r, 2)$, the inductive hypothesis implies there is $A_{i+1} \subseteq A_i - \{a_i\}$ with $|A_{i+1}| \geq N(i+1)$ and $A_{i+1}^{(r)} \subseteq Q_1$ or $A_{i+1}^{(r)} \subseteq Q_2$. Let a_{i+1} be any element of A_{i+1} . This finishes the construction of the A_i 's and a_i 's. Let $A = \{a_1, \dots, a_{2M}\}$, and note that for each $i \in \{1, \dots, 2M\}$,

either (1) $\{a_i, a_{i(1)}, \dots, a_{i(r)}\} \in P_1$ whenever $i < i(1) < \dots < i(r) \leq 2M$,

or (2) $\{a_i, a_{i(1)}, \dots, a_{i(r)}\} \in P_2$ whenever $i < i(1) < \dots < i(r) \leq 2M$.

Let $Y_1 := \{a_i : (1) \text{ holds}\}$ and $Y_2 := \{a_i : (2) \text{ holds}\}$. Then $A = Y_1 \cup Y_2$, and either $|Y_1| \geq M$ or $|Y_2| \geq M$, and clearly $Y_1^{(r+1)} \subseteq P_1$ and $Y_2^{(r+1)} \subseteq P_2$. So either $Y = Y_1$ or $Y = Y_2$ has the required property. $\square$

(3.5) Ramsey's theorem is intimately related to the model-theoretic notion of indiscernibility, which we need here only in a very simple form, as follows.

DEFINITIONS. Let X be an infinite set. Given a relation $A \subseteq X^r$ we say that a finite sequence $x_1, \dots, x_M$ in X is **A-indiscernible** if

$$(x_{i(1)}, \dots, x_{i(r)}) \in A \Leftrightarrow (x_{j(1)}, \dots, x_{j(r)}) \in A.$$

whenever $1 \leq i(1) < \dots < i(r) \leq M$ and $1 \leq j(1) < \dots < j(r) \leq M$. Let $\mathcal{A}$ be a collection of relations on X , that is, each element of $\mathcal{A}$ is a set $A \subseteq X^r$, with $r \in \mathbb{N}$ depending on A . Then a sequence $x_1, \dots, x_M$ in X is called **$\mathcal{A}$ -indiscernible** if it is A -indiscernible for each $A \in \mathcal{A}$. A **subsequence** of a sequence $x_1, \dots, x_N$ is by definition a sequence $x_{i(1)}, \dots, x_{i(M)}$ with $1 \leq i(1) < \dots < i(M) \leq N$.

(3.6) COROLLARY. *Let X be an infinite set and $\mathcal{A}$ a finite collection of relations on X . Then, given any positive integer M there is a positive integer N so large that each sequence in X of length N contains an $\mathcal{A}$ -indiscernible subsequence of length M .*

PROOF. By an induction on $|\mathcal{A}|$ it suffices to prove this when $\mathcal{A}$ consists of just one relation $A \subseteq X^r$, $r > 0$. Let $x_1, \dots, x_N$ be a sequence in X of length N , with $N = N(M, r, 2)$, and write $\{1, \dots, N\}^{(r)} = P_1 \cup P_2$ with

$$P_1 := \{\{i(1), \dots, i(r)\} : 1 \leq i(1) < \dots < i(r) \leq N, (x_{i(1)}, \dots, x_{i(r)}) \in A\},$$

$$P_2 := \{\{i(1), \dots, i(r)\} : 1 \leq i(1) < \dots < i(r) \leq N, (x_{i(1)}, \dots, x_{i(r)}) \notin A\}.$$

By Ramsey's theorem there is $\{i(1), \dots, i(M)\} \subseteq \{1, \dots, N\}$, with $i(1) < \dots < i(M)$, such that $\{i(1), \dots, i(M)\}^{(r)} \subseteq P_1$ or $\{i(1), \dots, i(M)\}^{(r)} \subseteq P_2$. Then the subsequence $x_{i(1)}, \dots, x_{i(M)}$ of $x_1, \dots, x_N$ is clearly A -indiscernible. $\square$

(3.7) For the rest of this section we fix infinite sets X and Y and a binary relation $\Phi \subseteq X \times Y$.

(3.8) LEMMA. *Suppose Φ is independent, and $\mathcal{A}$ is a finite collection of relations on X . Then there are for each $M \in \mathbb{N}$ an $\mathcal{A}$ -indiscernible sequence $a_1, \dots, a_M$ in X and an element $b \in Y$ such that for all $m \in \{1, \dots, M\}$, $(a_m, b) \in \Phi \Leftrightarrow m$ is even.*

PROOF. Fix a natural number M . By (3.6) we can find a natural number N so large that each sequence in X of length N contains an $\mathcal{A}$ -indiscernible subsequence of length M . As Φ is independent there are elements $x(i) \in X$ for $1 \leq i \leq N$ such that the sets $\Phi_{x(1)}, \dots, \Phi_{x(N)}$ are independent; so for each set $w \subseteq \{1, \dots, N\}$ the intersection

$$(1) \quad \left(\bigcap_{i \in w} \Phi_{x(i)} \right) \cap \left(\bigcap_{i \notin w} (\Phi_{x(i)})^{-1} \right)$$

is nonempty.

Fix an $\mathcal{A}$ -indiscernible subsequence $x(i_1), \dots, x(i_M)$ of length M of $(x(i))_{1 \leq i \leq N}$, and take for b an element of the intersection (1) for $w := \{i_{2m} : 1 \leq m \leq M/2\}$. Then the lemma above holds for the sequence $a_1, \dots, a_M$ where $a_m := x(i_m)$. $\square$

(3.9) Next we introduce certain relations that are in some sense “definable” from Φ . Let the variables $x_1, \dots, x_M$ range over X and the variable y over Y .

Given a positive integer M and a set $u \subseteq \{1, \dots, M\}$ we put

$$\Phi_u(x_1, \dots, x_M; y) := \left(\bigwedge_{i \in u} \Phi(x_i, y) \right) \& \left(\bigwedge_{i \notin u} \neg \Phi(x_i, y) \right),$$

a formula defining a subset Φ_u of $X^M \times Y$; the image of Φ_u under the projection map $X^M \times Y \rightarrow X^M$ is defined by the formula $\exists y \Phi_u(x_1, \dots, x_M; y)$, and here we simply regard this last formula as a convenient notation for the subset of X^M it defines. So

$$\mathcal{A}_{\Phi, M} := \{\exists y \Phi_u(x_1, \dots, x_M; y) : u \subseteq \{1, \dots, M\}\}$$

is simply a finite collection of M -ary relations on X .

The following is a converse to the previous lemma.

(3.10) LEMMA. *Let $a(1), \dots, a(N)$ be a sequence in X that is $\mathcal{A}_{\Phi, M}$ -indiscernible, $N \geq 2M$, let $a(i_1), \dots, a(i_{2M})$ be a subsequence, and let $b \in Y$ be such that for all $m \in \{1, \dots, 2M\}$, $(a(i_m), b) \in \Phi \Leftrightarrow m$ is even. Then $D(\Phi) > M$.*

PROOF. It suffices to show that $\Phi_{a(1)}, \dots, \Phi_{a(M)}$ are independent. Note first that $\Phi_E(a(i_1), \dots, a(i_{2M}); b)$ holds for $E :=$ set of even integers in $\{1, \dots, 2M\}$. Given any set $u \subseteq \{1, \dots, M\}$, take a sequence $1 \leq k(1) < \dots < k(M) \leq 2M$ such that $k(i)$ is even for $i \in u$ and $k(i)$ is odd for $i \notin u$. Then clearly

$$\Phi_u(a(i_{k(1)}), \dots, a(i_{k(M)}); b)$$

holds. Hence $\exists y \Phi_u(a(i_{k(1)}), \dots, a(i_{k(M)}); y)$ holds, and thus

$$\exists y \Phi_u(a(1), \dots, a(M); y)$$

holds by indiscernability. Since $u \subseteq \{1, \dots, M\}$ was arbitrary this shows that $\Phi_{a(1)}, \dots, \Phi_{a(M)}$ are independent. $\square$

(3.11) Next we assume also that the set Y is given as a cartesian product $Y = Y_1 \times Y_2$, with both Y_1 and Y_2 infinite. Let the variable y_1 range over Y_1 and y_2 over Y_2 . Consider now the formula $\Phi(x; y_1, y_2)$ as defining a relation $\Phi^* \subseteq (X \times Y_1) \times Y_2$. (This is just Φ if we identify $(X \times Y_1) \times Y_2$ with $X \times (Y_1 \times Y_2)$ in the usual way.) The relation Φ^* parametrizes a collection of subsets of Y_2 with index set $X \times Y_1$, and so it makes sense to say that Φ^* is dependent.

Given a positive integer M and a set $u \subseteq \{1, \dots, M\}$ we introduce a formula

$$\Gamma_{\Phi, u}(x_1, \dots, x_M; y_1) := \exists y_2 \Phi_u(x_1, \dots, x_M; y_1, y_2).$$

This formula $\Gamma_{\Phi, u}$ defines a subset of $X^M \times Y_1$, and we will simply regard $\Gamma_{\Phi, u}$ as a notation for this subset. So $\Gamma_{\Phi, u}$ parametrizes a collection of subsets of Y_1 with index set X^M . So it makes sense to say that $\Gamma_{\Phi, u}$ is dependent. With these notations we have

(3.12) THEOREM. *Suppose there are positive integers M and N such that $D(\Phi^*) \leq M$ and $D(\Gamma_{\Phi, u}) \leq N$ for all $u \subseteq \{1, \dots, M\}$. Then Φ is dependent.*

REMARK. The constructive nature of the proof below and of the arguments above would in principle allow us to extract a (huge) bound $D(\Phi) \leq B(M, N)$, for some

explicit function B of M and N . The construction of this function B would involve the bounds in Ramsey's theorem.

PROOF. We introduce a formula $\Psi_{u,v}$ for $u \subseteq \{1, \dots, M\}$ and $v \subseteq \{1, \dots, N\}$:

$$\Psi_{u,v}(x^1, \dots, x^N) := \exists y_1 \left(\bigwedge_{j \in v} \Gamma_{\Phi,u}(x^j; y_1) \ \& \ \bigwedge_{j \notin v} \neg \Gamma_{\Phi,u}(x^j; y_1) \right),$$

where each $x^j = (x^{j1}, \dots, x^{jM})$ is a variable ranging over X^M . So $\Psi_{u,v}$ defines a subset of $(X^M)^N = X^{MN}$. Actually $\Psi_{u,v}$ is just $\exists y_1 (\Gamma_{\Phi,u})_v(x^1, \dots, x^N; y_1)$ in the notation of (3.9) above. Put $\mathcal{A}_u := \{\Psi_{u,v} : v \subseteq \{1, \dots, N\}\}$ and

$$\mathcal{A} := \{\Psi_{u,v} : u \subseteq \{1, \dots, M\}, v \subseteq \{1, \dots, N\}\},$$

so $\mathcal{A}$ is the union of the $\mathcal{A}_u$'s.

Suppose Φ is independent. (We shall derive a contradiction.) Put

$$K := (2N)^{2^M} \cdot 2M.$$

By lemma (3.8) there are an $\mathcal{A}$ -indiscernible sequence $a_1, \dots, a_K$ in X and an element $b = (b_1, b_2) \in Y_1 \times Y_2$ such that $\Phi(a_k; b)$ holds if and only if k is even. Since $D(\Phi^*) \leq M$ there is by lemma (3.10) no interval J of length $2M$ contained in $\{1, \dots, K\}$ so that the sequence $(a_k, b_1)_{k \in J}$ is $\mathcal{A}_{\Phi^*, M}$ -indiscernible.

Each relation in $\mathcal{A}_{\Phi^*, M}$ is defined by a formula of the form

$$\exists y_2 \Phi_u^*((x_1, y_{11}), \dots, (x_M, y_{1M}); y_2)$$

with $u \subseteq \{1, \dots, M\}$ and each variable (x_i, y_{1i}) ranging over $X \times Y_1$. For $(a_k, b_1)_{k \in J}$ to be indiscernible with respect to this relation is equivalent to $(a_k)_{k \in J}$ being $\Gamma_{\Phi,u}(x_1, \dots, x_M; b_1)$ -indiscernible. (Here $\Gamma_{\Phi,u}(x_1, \dots, x_M; b_1)$ stands for the subset of X^M defined by it.) So there is no interval J of length $2M$ contained in $\{1, \dots, K\}$ such that $(a_k)_{k \in J}$ is $\Gamma_{\Phi,u}(x_1, \dots, x_M; b_1)$ -indiscernible for all $u \subseteq \{1, \dots, M\}$.

CLAIM. Let P and Q be positive integers with $Q \geq 2NP$, let $I = \{k : i_0 \leq k < i_0 + Q\}$ be an interval of length Q contained in $\{1, \dots, K\}$, and let $u \subseteq \{1, \dots, M\}$. Then I has a subinterval $J := \{k : j_0 \leq k < j_0 + P\}$ of length P , such that the sequence $(a_k)_{k \in J}$ is $\Gamma_{\Phi,u}(x_1, \dots, x_M; b_1)$ -indiscernible.

Note that if the claim holds, then starting with $Q := K$ and $P := Q/2N$, and successively applying the claim to each of the 2^M subsets $u \subseteq \{1, \dots, M\}$ we obtain in the end an interval J of length $2M$ contained in $\{1, \dots, K\}$ so that the sequence $(a_k)_{k \in J}$ is $\Gamma_{\Phi,u}(x_1, \dots, x_M; b_1)$ -indiscernible for all $u \subseteq \{1, \dots, M\}$, and we have a contradiction as desired. So all that remains is to prove the claim.

PROOF OF CLAIM. Assume there is no such interval J . Now, given any j with $0 \leq j < 2N$ the interval $J(j) := \{k : i_0 + jP \leq k < i_0 + (j+1)P\}$ is of length P and contained in I , so in particular $(a_k)_{k \in J(j)}$ is not $\Gamma_{\Phi,u}(x_1, \dots, x_M; b_1)$ -indiscernible. So for some strictly increasing sequence of length M in $J(j)$ the corresponding sequence of a 's does not satisfy $\Gamma_{\Phi,u}(x_1, \dots, x_M; b_1)$. When j is even we choose a strictly increasing sequence $k(j, 1) < \dots < k(j, M)$ in $J(j)$ of the first kind, while for odd j we choose $k(j, 1) < \dots < k(j, M)$ in $J(j)$ of the second kind. Hence $\Gamma_{\Phi,u}(a_{k(j,1)}, \dots, a_{k(j,M)}; b_1)$ holds if and only if j is even, for $0 \leq j < 2N$. Put $a_j^* := (a_{k(j,1)}, \dots, a_{k(j,M)}) \in X^M$ for $0 \leq j < 2N$. Since $(a_k)_{k \in I}$ is an $\mathcal{A}$ -indiscernible sequence, the order in which the a_k 's appear within an "increasing" sequence of a_j^* 's implies that $(a_j^*)_{0 \leq j < 2N}$ is $\mathcal{A}_u$ -indiscernible. But $\Gamma_{\Phi,u}(a_j^*; b_1)$ holds if and only if j is even, which by lemma (3.10) contradicts $D(\Gamma_{\Phi,u}) \leq N$. $\square$

(3.13) We call attention to the fact that for dependence of Φ we need only dependence of the auxiliary relations Φ^* and $\Gamma_{\Phi,u}$, which are defined purely in terms of Φ ; also, the definitions of Φ^* and $\Gamma_{\Phi,u}$ in terms of Φ do not involve quantifiers over X .

(3.14) PROOF OF THEOREM (3.1). We are given an infinite model-theoretic structure $\mathcal{R} = (R, \dots)$ such that each definable relation $\Phi \subseteq R^p \times R$, for each $p > 0$, is dependent. We have to show that all definable relations $\Phi \subseteq R^p \times R^q$ are dependent, for all $p, q > 0$. The case $q = 1$ holds by assumption. Assume inductively that the desired property holds for a certain $q > 0$, and let $\Phi \subseteq R^p \times R^{q+1}$ be definable, $p > 0$. Now set $X := R^p$, $Y_1 := R^q$, and $Y_2 := R$, $Y := Y_1 \times Y_2 = R^{q+1}$. Then $\Phi \subseteq X \times Y$, so we are in the setting of (3.11) above. By the hypothesis of the theorem the (definable) relation $\Phi^* \subseteq R^{p+q} \times R$ is dependent, say $D(\Phi^*) \leq M$, for a certain positive integer M . Since clearly all relations $\Gamma_{\Phi,u} \subseteq R^{M+p} \times R^q$ with $u \subseteq \{1, \dots, M\}$ are definable, the inductive assumption implies there is some positive integer N such that $D(\Gamma_{\Phi,u}) \leq N$ for all $u \subseteq \{1, \dots, M\}$. Then theorem (3.12) tells us that Φ is dependent, as desired. $\square$

(3.15) COROLLARY. *Let $(R, <, \mathcal{S})$ be an o-minimal structure and $\Phi \subseteq R^p \times R^q$ ($p, q > 0$) a definable relation. Then Φ is dependent. In particular, there is $d = d(\Phi) \in \mathbb{N}$ such that for all sufficiently large n each n -element set $F \subseteq R^q$ has at most n^d subsets of the form $\Phi_x \cap F$ with $x \in R^p$.*

PROOF. From (2.7) it follows that $(R, <, \mathcal{S})$ satisfies the hypothesis of theorem (3.1). Hence Φ is dependent. Thus $\Phi_X = \{\Phi_x : x \in R^p\}$ is a VC-class, by (2.11). $\square$

Notes and comments

The combinatorial dichotomy of Section 1 is due to Shelah [53], and independently to Vapnik and Chervonenkis [62]. In [53] this had to do with the model-theoretic “independence property”, and in this connection Shelah also proved (3.1) by a curious set-theoretic argument.

In [62] the motivation came from probability theory: if $\mathcal{C}$ is a VC-class of events in a probability space X , then the estimates in Bernoulli’s theorem for the events in $\mathcal{C}$ are uniform in a certain technical sense. See also Dudley [24] for more on this.

That o-minimal structures do not have the independence property was shown by Pillay and Steinhorn [49]. The combinatorial interpretation in terms of the VC-property was provided by Laskowski [38] who also gave the purely combinatorial proof of (3.1) in Section 3. Wilkie (unpublished) gave another proof of (3.1) for o-minimal structures using their special properties. Somewhat earlier Stengle and Yukich [58] had shown (as in (2.13) and (2.14) above) that semialgebraic collections have the VC-property.

For recent applications to the study of neural networks, see Macintyre and Sontag [41] and Sontag [56].

